FIITJEE

# JEE (Advanced)-2018

ALL INDIA TEST SERIES

## OPEN TEST

## Paper 1

**Time Allotted: 3 Hours** **Maximum Marks: 183**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

## INSTRUCTIONS

### A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

### B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

### C. Marking Scheme For All Three Parts.

1. ***Section-A (01– 07, 19 – 25, 37 - 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   ***Section-A (08 – 13, 26 – 31, 44 - 49)*** *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S).*
   *Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (14 – 18, 32 – 36, 50 - 54)*** *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

| | |
|---|---|
| **Name of the Candidate** | |
| **Enrolment No.** | |

## Useful Data

### PHYSICS

| | |
|---|---|
| Acceleration due to gravity | $g = 10\ m/s^2$ |
| Planck constant | $h = 6.6 \times 10^{-34}$ J-s |
| Charge of electron | $e = 1.6 \times 10^{-19}$ C |
| Mass of electron | $m_e = 9.1 \times 10^{-31}$ kg |
| Permittivity of free space | $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$ |
| Density of water | $\rho_{water} = 10^3\ kg/m^3$ |
| Atmospheric pressure | $P_a = 10^5\ N/m^2$ |
| Gas constant | $R = 8.314\ J\ K^{-1}\ mol^{-1}$ |

### CHEMISTRY

| | | | |
|---|---|---|---|
| Gas Constant | R | = | $8.314\ J\ K^{-1}\ mol^{-1}$ |
| | | = | $0.0821$ Lit atm $K^{-1}\ mol^{-1}$ |
| | | = | $1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$ |
| Avogadro's Number | $N_a$ | = | $6.023 \times 10^{23}$ |
| Planck's constant | h | = | $6.625 \times 10^{-34}$ J·s |
| | | = | $6.625 \times 10^{-27}$ erg·s |
| 1 Faraday | | = | 96500 coulomb |
| 1 calorie | | = | 4.2 joule |
| 1 amu | | = | $1.66 \times 10^{-27}$ kg |
| 1 eV | | = | $1.6 \times 10^{-19}$ J |

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

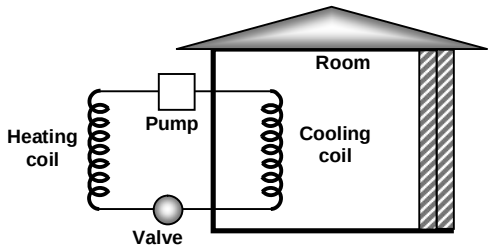
Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

# *Physics* PART – I

## SECTION – A
### (One or More than one correct type)

This section contains **SEVEN** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A room air conditioner is modelled as a heat engine run in reverse. An amount of heat $Q_{in}$ is absorbed from the room at a temperature $T_L$ by the cooling coils which contains a working gas; this gas is compressed adiabatically to a temperature $T_H$; then the gas is compressed isothermally in a coil outside of the house, giving off an amount of heat $Q_{out}$; the gas expands adiabatically back to a temperature $T_L$ and the cycle is repeated. An amount of energy W is input into the system in every cycle through an electric pump. This model describes the air conditioner with the best possible efficiency.

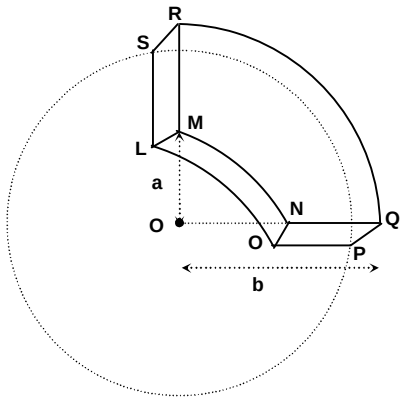



The outside air temperature is $T_H$ and the inside air temperature is $T_L$. The air conditioner unit consumes electric power P. Assume that the air is dry sufficiently so that no condensation of water occurs in the cooling coils of the air conditioner. Which of the following may be the rate at which heat is removed from the room?

(A) $\frac{P}{2}\left(\frac{T_L}{T_H - T_L}\right)$ (B) $\frac{3P}{4}\left(\frac{T_L}{T_H - T_L}\right)$

(C) $P\left(\frac{T_L}{T_H - T_L}\right)$ (D) $\frac{3P}{2}\left(\frac{T_L}{T_H - T_L}\right)$

2. A conducting medium is shaped in a form of a quarter of an annulus of radii b and a (b > a) and thickness t. Specific resistance of the medium is $\rho$. The correct option(s) is/are




(A) Resistance between faces LMNO and PQRS is $\frac{2\rho}{\pi t}\ell n\frac{b}{a}$.

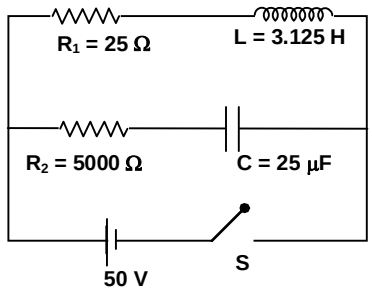
(B) Resistance between faces LMRS and ONQP is $\frac{\pi\rho}{2t\ell n\left(\frac{b}{a}\right)}$.

(C) Resistance between faces MRQN and LSPO is $\frac{4t\rho}{b^2 - a^2}$.

(D) Between faces LMNO & PQRS and LMRS & ONQP, resistance is same.

***Space for Rough work***

3. Two resistors, one inductor and a capacitor are connected in a circuit with a battery and switch as shown in figure. The charge on capacitor and current in inductor is zero at t = 0. Now switch is closed at t = 0, then choose the correct option(s).



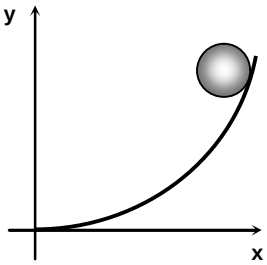

(A) The ratio of maximum current in the branch of inductor to the branch of capacitor is 200.

(B) At $t = \frac{\ell n 2}{8}$, the current in the branch of capacitor is 200 times the current in the branch of inductor.

(C) At $t = \frac{1}{8}\ell n\left(\frac{5}{4}\right)$, the potential drop across inductor is 4 times the potential drop across capacitor.

(D) The potential drop across inductor is maximum at t = 0.

4. A small disc of mass m is released on a parabolic curve in a vertical plane such that gravity acts along negative y-axis. The equation of parabolic curve is $x^2 = \frac{2a}{\sqrt{3}}y$, where 'a' is a positive constant. Frictional force between disc and curve are sufficient for pure rolling. When disc is reached at x = a then choose the correct option(s).



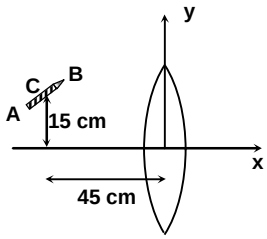

(A) acceleration of disc along the trajectory is $\sqrt{3}g$.

(B) acceleration of disc along the trajectory is $\frac{g}{\sqrt{3}}$.

(C) frictional force between disc and curve is $\frac{mg}{2\sqrt{3}}$.

(D) frictional force between disc and curve is $\frac{mg}{\sqrt{3}}$.

5. A $10\sqrt{2}$ cm long pencil is placed at an angle of 45° with optical axis. Its centre is 15 cm above the optics axis and 45.0 cm from a lens with 20.0 cm focal length as shown in the figure. Assume that the diameter of the lens is large enough for the paraxial approximation to be valid. For approximate position of image choose the correct option(s).




(A) The location of image of point A is (30 cm, −6.10 cm)

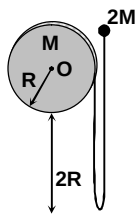
(B) The location of image of point B is (40.0 cm, −20.0 cm)

(C) The location of image of point C is (36.0 cm, −12.0 cm)

(D) The location of image of point A is (33.3 cm, −6.6 cm)

***Space for Rough work***

6. A light string is wrapped on a cylindrical shell and a fraction of length of string is unwrapped. A particle of mass 2M is attached on another end of string as shown. The system is kept in a vertical plane and cylinder can freely rotate about the axis of cylinder. Particle is released as shown in figure. Assuming thee is no slipping between cylinder and string. Choose the correct option(s).



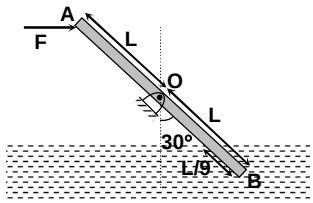

(A) The angular velocity of cylinder just after string taut is $\frac{8}{3}\sqrt{\frac{g}{R}}$.

(B) The angular velocity of cylinder just after string taut is $\frac{16}{3}\sqrt{\frac{g}{R}}$.

(C) Velocity of particle just after string taut is $\frac{8}{3}\sqrt{gR}$.

(D) Velocity of particle just after string taut is $\frac{16}{3}\sqrt{gR}$.

7. A rod of length 2L and cross section area $A_0$ is kept in a liquid of density $\rho$ and pivoted at point O in a vertical plane. The rod makes angle 30° with the vertical. The mass density of rod is linearly increases from O on both side as given by equation $\lambda = \lambda_0\left(1+\frac{x}{L}\right)$, where $\lambda_0$ is constant. Rod is held in vertical plane by applying a force at end A in horizontal direction. Then




(A) Centre of mass of rod OA is at a distance $\frac{4L}{9}$ from point A.

(B) Mass of rod AB is $\frac{3\lambda_0 L}{2}$.

(C) Applied force at end A is $\frac{17}{162\sqrt{3}}LA_0\rho g$.

(D) Buoyant force will act at the centre of mass of rod inside the liquid.

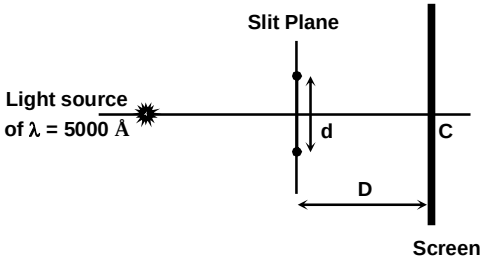
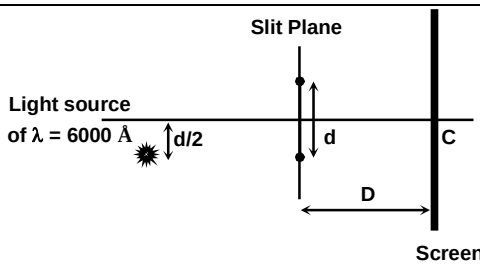
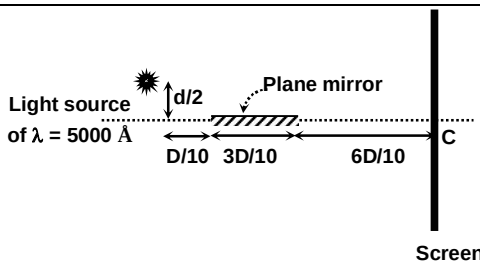
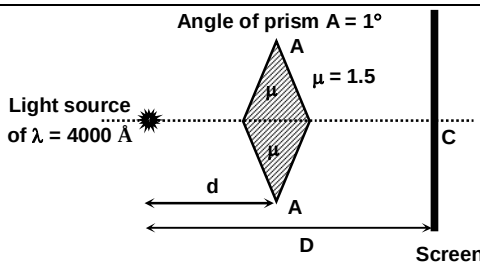
***Space for Rough work***

**(Matching type - Single Correct Option)**

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

**Answer questions 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.**

Column 1 shows four different setups for observing interference on a screen for different values of D (in column 2) and d (in column 3). D and d are mention in the diagram.

| Column 1 (Setup) | | Column 2 (D) | | Column 3 (d) | |
|---|---|---|---|---|---|
| (I) |  | (i) | 1/2 m | (P) | 2mm |
| (II) |  | (ii) | 2 m | (Q) | 1mm |
| (III) |  | (iii) | 3 m | (R) | 4 mm |
| (IV) |  | (iv) | 3/2 m | (S) | $\frac{9}{\pi}$ cm |

***Space for Rough work***

8. For which of combination fringe width is maximum?
(A) (I) (iv) (S) (B) (III) (i) (P)
(C) (II) (iii) (Q) (D) (IV) (ii) (R)

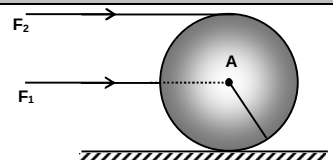
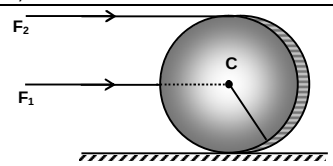
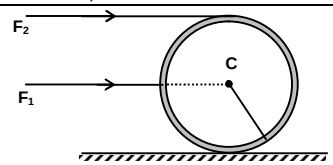
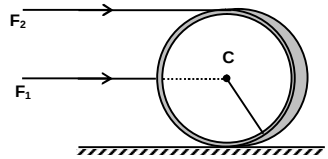
9. For which of the combination $\frac{I_C}{I_{max}}$ is minimum, where $I_C$ is the intensity at point C and $I_{max}$ is the maximum intensity on the screen.
(A) (II) (iv) (P) (B) (II) (i) (Q)
(C) (III) (ii) (R) (D) (IV) (iii) (S)

10. For which of the combination no. of fringes on the screen is minimum?
(A) (I) (iii) (P) (B) (III)(i) (R)
(C) (II) (ii) (R) (D) (IV) (iii) (S)

***Space for Rough work***

**Answer questions 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.**

A sphere, a cylinder, a spherical shell and a cylindrical shell are kept on a horizontal surface. Two forces $F_1$ and $F_2$ are acting on each object $F_1$ passes through centre of mass $F_2$ acts on the top most point. All objects starts from rest on a smooth horizontal surface.

| Column 1 | | Column 2 | | Column 3 | |
|---|---|---|---|---|---|
| (I) |   Uniform solid sphere of mass m, radius R | (i) | $F_1 = F_2 = F$ | (P) | Object will roll with out sliding |
| (II) |   Uniform solid cylinder of mass m, radius R | (ii) | $F_2 = 2F_1 = 2F$ | (Q) | Object will roll on the horizontal surface with $a_{cm} < R\alpha$ |
| (III) |   Uniform thin spherical shell of mass m, radius R | (iii) | $F_1 = 0, F_2 = F$ | (R) | Object will roll on the horizontal surface with $2R\alpha > a_{cm} > R\alpha$ |
| (IV) |   Uniform thin cylindrical shell of mass m, radius R | (iv) | $F_1 = \frac{3F_2}{2} = \frac{3F}{2}$ | (S) | Object will roll on the horizontal surface with $a_{cm} \geq 2R\alpha$ |

11. Mark the correct statement.
(A) (I) (i) (P) (B) (II)(ii) (Q)
(C) (III) (ii) (R) (D) (IV) (iii) (S)

12. Mark the incorrect statement.
(A) (I) (ii) (Q) (B) (II)(i) (P)
(C) (III) (iv) (R) (D) (IV) (iii) (S)

13. From the given combination the kinetic energy for time t will be maximum.
(A) (I) (ii) (Q) (B) (II)(iv) (R)
(C) (III) (iii) (P) (D) (IV) (i) (S)

***Space for Rough work***

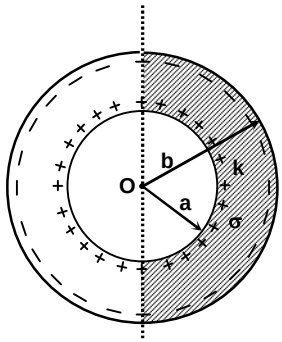
## SECTION – C
### (Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. A particle having specific charge $\sigma$ is projected in xy plane with a speed v. There exists a uniform magnetic field in z-direction having a fixed magnitude $B_0$. The field is made to reverse its direction after every interval of $\frac{2\pi}{\sigma B_0}$. Calculate the maximum separation (in m) between two positions of the particle during its course of motion. (given $\frac{v}{\sigma B_0} = 2$ metre) (neglect any other force including gravity throughout the motion)

15. The half lives of two longest lived radioactive isotopes of phosphorus $P^{32}$ and $P^{33}$ are 14 days and 28 days respectively. A sample has been prepared by mixing the two isotopes in the ratio of 4 : 1 of their atoms. Initial activity of the mixed sample is 90 m curie. Find the activity (in m curie) of the sample after 80 days. (Approximately to nearest integer) Take $\ell n2 = 0.7$ and $e^{-2} = 0.14$

16. Half portion of a spherical capacitor is filled with a dielectric of dielectric constant k = 2 conductivity $\sigma$. The charge given to spherical capacitor is $Q_0$. Due to the conductivity of dielectric charge leaks the time constant for the discharge circuit is x time of $\frac{\varepsilon_0}{\sigma}$. Find the value of x.




17. Eight identical droplets of water were falling in air with constant velocity 2 cm/s. The initial acceleration of each droplet was '$a_1$' at the time when they starts falling. At a time 't' these droplets coalesce to form a big drop due to pressure difference developed, and just after the formation of combined drop its acceleration is found '$a_2$'. The ratio of $\frac{6a_1}{a_2}$ is

18. A pipe 20 cm long is closed at one end which harmonic mode of the pipe is resonantly exited by a source of 1237.5 Hz. Sound velocity in air 330 m/s.

***Space for Rough work***

# *Chemistry* PART – II

## SECTION – A
### (One or More than one correct type)

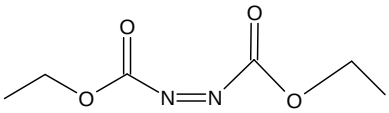
This section contains **SEVEN** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Choose the correct statements:
(A) temperature, enthalpy and entropy are state functions
(B) for reversible and irreversible both isothermal expansion of an ideal gas, change in internal energy is zero
(C) for a reaction in which $\Delta n_g = 0$, entropy change is not always zero
(D) the entropy change associated with reversible isothermal expansion of an ideal gas is equal to $2.303\ R \log_{10} \frac{P_1}{P_2}$

20. Carefully study the following 2 cases and choose the correct statements followed?
**Case # I.** All H atoms in ethene are substituted differently without significantly distorting the original geometry.
**Case # II.** All H atoms in ethane are substituted differently without significantly distorting the original geometry.
Choose the correct statement(s) from the following:

(A) Diastereomers will be produced only in Case I, not in Case II.
(B) Enantiomers will be produced only in Case II, not in Case I.
(C) Mixing equal weights of one dextro and one laevo molecule under Case II will necessarily yield an optically inactive mixture.
(D) Total number of space isomers possible under both cases = 86.

21. DEAD is an organic molecule standing for **D**i **E**thyl **A**zo **D**icarboxylate having the following structure:



Which of the following gas(es) will be evolved when DEAD is strongly heated with excess conc. $H_2SO_4$
(A) $N_2$ (B) $C_2H_4$
(C) CO (D) $CO_2$

22. Inorganic benzene surpasses organic benzene in which property:
(A) melting point (B) edge length of its hexagon
(C) molar mass (D) total number of electrons in the molecule

***Space for Rough work***

23. If equatorial plane in $PCl_5$ molecule is the X—Y plane, the orbitals hybridizing to produce axial bonds will be:

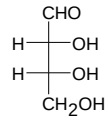
(A) $p_z$ (B) $d_{yz}$

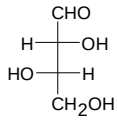
(C) $p_y$ (D) $d_{z^2}$

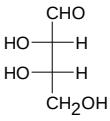
24. Which of the following changes can be experimentally realized in the laboratory:

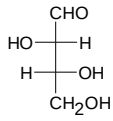
(A) combustion of Mg ribbon in $CO_2$
(B) reduction of $FeCl_3$ (aq.) on adding Zn granules
(C) dry heating of hydrated $MgCl_2$ to get anhydrous $MgCl_2$
(D) evolution of brown gas on heating $LiNO_3$

25. An optically active aldotetrose which retains its optical activity with aqueous $Br_2$ but loses it with conc. $HNO_3$ has the following structure:

(A) 


(B) 


(C) 


(D) 


***Space for Rough work***

**(Matching type - Single Correct Option)**

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

**Answer Q. 26, Q. 27 and Q. 28 by appropriately matching the information given in the three columns of the following table.**

| Column 1 (Polymer) | | Column 2 (Monomer) | | Column 3 | |
|---|---|---|---|---|---|
| (I) | Polystyrene | (i) | Caprolactum | (P) | Resonance stabilization |
| (II) | Poly Vinyl Chloride | (ii) | $PhCH = CH_2$ | (Q) | 1, 4 addition of HBr (mainly) at $40^oC$ |
| (III) | Natural rubber | (iii) | $CH_2 = CHCl$ | (R) | 1, 2 addtion of HBr at room temp. (mainly and as per traditional Markonikoff's rule) |
| (IV) | Nylon-6 | (iv) | $CH_2 = C(CH_3) – CH = CH_2$ | (S) | $CO_2(g)$ evolution with $KMnO_4/H^+$. |

26. Which of the following combination is CORRECT?
(A) (III) (ii) (R) (B) (II)(i) (P)
(C) (IV) (i) (P) (D) (I) (iii) (S)

27. Which of the following combination is INCORRECT?
(A) (I) (ii) (P) (B) (I)(ii) (R)
(C) (III) (iv) (P) (D) (II) (iv) (R)

28. Which of the following combination is CORRECT?
(A) (II) (ii) (P) (B) (II)(iii) (R)
(C) (III) (iii) (P) (D) (IV) (ii) (P)

***Space for Rough work***

**Answer Q. 29, Q. 30 and Q. 31 by appropriately matching the information given in the three columns of the following table.**

| Column 1 | | Column 2 | | Column 3 | |
|---|---|---|---|---|---|
| (I) | Peptization | (i) | Colloid preparation technique | (P) | Sky is blue |
| (II) | Delta formation | (ii) | Colloid destabilization | (Q) | Gold number |
| (III) | Tyndall effect | (iii) | Colloid stabilization | (R) | Coagulation |
| (IV) | Protection power of colloids | (iv) | Optical property of colloid | (S) | Adsorption of common ion |

29. Which of the following combination is CORRECT?
(A) (I) (i) (S) (B) (II)(i) (R)
(C) (III) (ii) (S) (D) (I) (ii) (S)

30. Which of the following combination is CORRECT?
(A) (II) (i) (S) (B) (II)(ii) (R)
(C) (III) (ii) (S) (D) (IV) (iii) (R)

31. Which of the following combination is INCORRECT?
(A) (III) (iv) (P) (B) (II) (ii) (R)
(C) (IV) (i) (P) (D) (I) (ii) (Q)

***Space for Rough work***

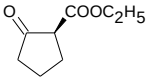
## SECTION – C
### (Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. Ammonium carbamate is heated at 227°C in a closed vessel of volume 8210 ml containing 0.2 mole of Argon gas. The manometer attached to the vessel shows a pressure of 1 atm. Find $K_p$ for the decomposition of ammonium carbamate.

$$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$$

33. How many gases/vapours are involved either as reactants or products in Serpeck's process used for concentration of bauxite.

34. 

When the above molecule is reduced with excess $LiAlH_4$ followed by dilute acid hydrolysis, how many organic products can be isolated by fractional distillation of product mixture.

35. The pH at which a 0.1 M $Al^{+3}$ solution is 99.99% precipitated is : ($K_{sp}$ $Al(OH)_3 = 1 \times 10^{-18}$ at 25°C)

36. 0.05 Faraday charge is passed in 100 ml, 0.1 M NaCl solution. Calculate the final pOH of the solution. Assume volume remains constant.

***Space for Rough work***

# *Mathematics* PART – III

## SECTION – A
### (One or More than one correct type)

This section contains **SEVEN** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let $P=\sum_{n=1}^{10} n\left\{\frac{1^2}{1+n}+\frac{2^2}{2+n}+.....+\frac{10^2}{10+n}\right\}$ then which of the following is/are correct?

(A) 2P is perfect square
(B) 2P is divisible by $11^2$
(C) Number of divisors of 2P is 9
(D) The remainder when 2P is divided by 6 is 1

38. Define a function $f(x)=\begin{cases}\sin x, \text{ if } x \text{ is irrational}\\ \tan x, \text{ if } x \text{ is rational}\end{cases}$. Then f(x) is

(A) not continuous at any point in the interval $(-\pi, \pi)$
(B) continuous at some point(s) , but not differentiable at any point in the interval $(-\pi, \pi)$
(C) continuous at some point(s) in the interval $(-\pi, \pi)$, but not differentiable at some of these point(s)
(D) continuous at some point(s) in the interval $(-\pi, \pi)$ and differentiable at all these point(s)

39. Let $A=\begin{bmatrix}1 & 2\\ -a^2 & a+2\end{bmatrix}$ and B, C are $2\times 2$ matrices such that $ABC=\begin{bmatrix}4 & 2\\ 1 & 3\end{bmatrix}$. Let $(BC)^T = [p_{ij}]_{2\times 2}$ and $p_{21} = -1$, then the value of a can be

(A) 0
(B) $-\frac{3}{2}$
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$

***Space for rough work***

40. A ray of light coming along the line $y = x + 1$ from positive x-direction strikes a concave mirror whose intersection with xy-plane is the parabola $(y - x)^2 = 16(x + y)$. Then which of the following is/are correct statement(s)

(A) reflected ray passes through (2, 2)
(B) slope of reflected ray is $\frac{47}{79}$
(C) reflected ray passes through (1, 1)
(D) slope of normal at point of contact is $\frac{7}{9}$

41. Let $L_1 : x + y = 1, z + 2y = 2$ and $L_2 : x - y = 1, z - 2y = 2$ be two straight lines, then which of the following is/are correct statement(s)?
(A) the lines $L_1$ and $L_2$ are coplanar
(B) the lines $L_1$ and $L_2$ are non-coplanar
(C) on the line $\vec{r} = 4\hat{j} + \lambda\left(\hat{i} + 2\hat{k}\right)$, $\lambda \in R$, there exist infinite number of points such that an infinite number of lines passing through each of these points and intersecting both the lines $L_1$ and $L_2$ can be drawn
(D) there exists a unique line passing through the point $A(1, \mu, 3)$ which intersect both lines $L_1$ and $L_2$ $\forall\ \mu \in R$.

42. Let $I = \int_1^2 \frac{e^x + e^{4/x}}{x} dx$, then which of the following is/are correct statement(s)?

(A) $2I = \int_1^4 \frac{e^x + e^{4/x}}{x} dx$
(B) $4I = \int_1^4 \frac{e^x + e^{4/x}}{x} dx$
(C) $I > (e^2 - e)$
(D) $I < (e^2 - e)$

43. Let $S_n = \sum_{r=1}^{n} \cot^{-1}\left(\frac{9 + r(r-1)}{3}\right)$, then which of the following is/are correct statement(s)?

(A) $S_n = \sin^{-1}\left(\frac{3}{\sqrt{n^2 - 9}}\right)$
(B) $S_n = \sin^{-1}\left(\frac{n}{\sqrt{n^2 + 9}}\right)$
(C) $S_n > \frac{n}{3}$
(D) $S_n < \frac{n}{3}$

***Space for rough work***

**(Matching type - Single Correct Option)**

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

**Answer Q.44, Q.45 and Q.46 by appropriately matching the information given in the three columns of the following table.**

Let $f(x) = \frac{(x-a)(x-b)}{(x-c)(x-d)}$ where a < b < c < d. Column-1: contains information about real roots of the equation; Column-2: contains information about limit of f(x); Column-3: contains information about increasing interval and decreasing interval of f(x) and f′(x) and information about local minimum value and local maximum value of f(x)

| Column–1 | Column–2 | Column–3 |
|---|---|---|
| (I) f(x) = 1 has exactly one real root | (i) $\lim_{x\to c^-} f(x) = -\infty$ | (P) f(x) is increasing in the interval (b, c) |
| (II) f(x) = 2 has exactly two real roots | (ii) $\lim_{x\to d^+} f(x) = \infty$ | (Q) f(x) is decreasing in the interval (d, ∞) |
| (III) f(x) = –10 has one real root | (iii) $\lim_{x\to d^-} f(x) = -\infty$ | (R) f′(x) is increasing in the interval (b, c) |
| (IV) f(x) = 10 has two real roots | (iv) $\lim_{x\to\infty} f(x) = 1$ | (S) local minimum value of f(x) is smaller than local maximum value of f(x) |

44. Which of the following is the only **CORRECT** combination?
(A) (I) (ii) (S) (B) (II) (iii) (Q)
(C) (I) (i) (R) (D) (III) (iv) (S)

45. Which of the following is the only **INCORRECT** combination?
(A) (II) (ii) (P) (B) (III) (i) (R)
(C) (IV) (iv) (P) (D) (I) (iii) (Q)

46. Which of the following is the only **CORRECT** combination?
(A) (III) (iv) (P) (B) (I) (ii) (R)
(C) (IV) (i) (S) (D) (II) (iv) (S)

***Space for rough work***

**Answer Q.47, Q.48 and Q.49 by appropriately matching the information given in the three columns of the following table.**

Let f(x) be a double differentiable function with continuous second derivative defined on the interval [0, 1] satisfying the equation $\int_0^x \sqrt{1-\left(f''(t)\right)^2}\,dt = \int_0^x f'(t)\,dt \ \forall\ x \in [0, 1]$ and f(0) = f′(0) = 0. Column-1: contains the information about interval of value of f(x); Column-2: contains information about value of f(x) ; Column-3: contains information about interval of value of definite integral of f(x); Match the following Column(s)

| Column–1 | Column–2 | Column–3 |
|---|---|---|
| (I) $f\left(\frac{1}{2}\right) < \frac{1}{8}$ | (i) $f\left(\frac{1}{3}\right) = \frac{1}{18}$ | (P) $\int_0^1 f(x)dx > 1$ |
| (II) $f\left(\frac{1}{2}\right) > \frac{1}{8}$ | (ii) $f\left(\frac{1}{4}\right) = \frac{1}{8}$ | (Q) $\int_0^1 f(x)dx > \frac{1}{6}$ |
| (III) $f\left(\frac{1}{3}\right) < \frac{1}{18}$ | (iii) $f\left(\frac{1}{6}\right) = 2\sin^2\left(\frac{1}{12}\right)$ | (R) $\int_0^1 f(x)dx < \frac{1}{6}$ |
| (IV) $f\left(\frac{1}{4}\right) < \frac{1}{4}$ | (iv) $f\left(\frac{1}{8}\right) = -2\sin^2\left(\frac{1}{16}\right)$ | (S) $\int_0^1 f(x)dx < \frac{1}{4}$ |

47. Which of the following is the only **CORRECT** combination?
(A) (I) (ii) (Q) (B) (II) (i) (S)
(C) (IV) (ii) (Q) (D) (III) (iii) (R)

48. Which of the following is the only **CORRECT** combination?
(A) (I) (iv) (Q) (B) (II) (iii) (S)
(C) (II) (iv) (P) (D) (IV) (iii) (S)

49. Which of the following is the only **CORRECT** combination?
(A) (III) (ii) (Q) (B) (IV) (ii) (P)
(C) (I) (iii) (R) (D) (III) (ii) (S)

***Space for rough work***

## SECTION – C
### (Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. The number of x in $(0, 2\pi)$ satisfying the equation $1 + \sin 2x = \sin x + \sin^2 x$ is _____

51. If $z_1$, $z_2$ are complex numbers, then the maximum value of $\frac{z_1\bar{z}_2 + \bar{z}_1 z_2 + z_1 z_2 + \bar{z}_1\bar{z}_2}{|z_1 z_2|}$ is equal to _____

52. Let PQ be a focal chord of $\frac{x^2}{9} + \frac{y^2}{4} = 1$ passing through one focus S and S′ be its other focus, then the value of semi-perimeter of $\Delta S'PQ$ is _____

53. Let $f(x) = ax^2 + bx + c$, where $a \neq 0$, a, b, c are integers and $f(1) = 1$, $6 < f(3) < 8$ and $18 < f(5) < 22$. Then the number of solutions of equation $f(x) = e^x$ is _____

54. Let $L = \lim_{x\to\infty}\left(\left(\frac{x+1}{x-1}\right)^x - e^2\right)x^2$, then the value of $\frac{9L}{2e^2}$ is equal to _____

***Space for rough work***